

Week 4: Inference – Variable Elimination and Belief Propagation

Overview, Resources and Notes

1 Overview

We are done with *modelling* (for now), and are moving on to *inference* this week. We start with *marginal inference*; that is, determining the marginal belief distribution over one RV given some evidence (in the form of observed values for RVs).

So far, we have performed marginal inference by first constructing the joint distribution of the model, then inserting the evidence, and lastly marginalising out the unwanted RVs. However, except for some trivial toy problems, explicitly constructing the joint distribution is prohibitively expensive. To address this issue, we now look at inference algorithms that exploit the structure of the model to avoid constructing the joint distribution, making inference possibly much cheaper.

This week focusses on the following two inference algorithms:

- **Variable elimination (VE)**: Instead of first multiplying all the factors together to form the joint distribution (a large table) and then marginalising out all the unwanted RVs (to form a small table), this algorithm alternates between multiplying factors and marginalising out RVs to keep the largest table that is formed in the process as small as possible.
- **Belief propagation (BP)**: This algorithm is very similar to VE, but has a very different intuition. In BP, the factors of the model is kept in a distributed form as nodes of a graph, and the nodes “communicate” with each other by sending messages across the edges of the graph. After messages have been passed throughout the whole graph, the desired belief distribution can be calculated by combining some of the messages.

For both these algorithms, we restrict ourselves for this week to models that can be represented as **factor trees** (i.e., factor graphs (FGs) that have a tree structure); we look at techniques for loopy FGs (i.e., FGs with loops) next week.

A bit of perspective: BP has some advantages over VE (see resources for details) and it has no significant drawbacks compared to VE. For this reason, VE is hardly used in practice these days; however, it provides the basis for BP, which is why we cover both algorithms.

2 Resources

Prescribed resources

Study the following resources before the lectures and practical on Thursday (see the notes further down to get an idea of what to focus on):

- **Barber** Section 5.1
- **Bishop** Sections 8.4 up to the end of Section 8.4.4
- **PGM Tutorial** introduction to Chapter 4 and Section 4.1

Supplementary resources (optional)

I also recommend that you study the following resources, if you have access to it:

- **Koller videos** in the Coursera course Probabilistic Graphical Models 2: Inference, Week 1:
 - *Overview: Conditional Probability Queries (15:22)*
 - *Overview: MAP Inference (9:47)*
 - *Variable Elimination: Variable Elimination Algorithm (16:17)*
 - *Variable Elimination: Complexity of Variable Elimination (12:48)*
 - *Variable Elimination: Graph-Based Perspective on Variable Elimination (15:25)*
 - *Variable Elimination: Finding Elimination Orderings (11:58)*

If you would like to read further, the following resources might be helpful: **Murphy** Sections 9.1 and 9.5; **Jordan** Sections 3.1, 4.1 and 4.2; **Kochenderfer** Sections 3.1 to 3.4; **Koller&Friedman** Chapter 9 (excluding Sections 9.5 and 9.6).

3 Notes

Comments on the resources

- The following concepts are central – make sure you understand them very well: the meaning of *marginal inference*; the variable elimination (VE) algorithm, including how to choose a good elimination ordering for a factor tree; why VE could be cheaper than first constructing the joint distribution and then marginalising out the unwanted RVs; the belief propagation (BP) algorithm in factor trees, including how messages are calculated and what an appropriate message-passing order is; the similarities and differences between VE and BP; how evidence is incorporated into the inference process for both VE and BP; how marginal (posterior) beliefs are calculated from messages in BP.
- Terminology: What is called *sum-product variable elimination* in the **PGM Tutorial** is what the other resources call *variable elimination* – it is different from the *sum-product algorithm* (see next point).
- More terminology: *Belief propagation (BP)* is a type of *message-passing algorithm*, and it is also called the *sum-product algorithm* and the *Shafer-Shenoy algorithm* in various resources.
- For BP in FGs, there are essentially two equivalent ways to extract the marginal posterior belief over a certain RV from messages in the FG:
 - At the *variable node* for the RV of interest: by multiplying together all the incoming messages to this node (as shown in **Barber** under the “Marginal” heading on p. 89).
 - At any *factor node* that contains the RV of interest: by multiplying all the incoming messages with the factor assigned to this node (as shown in **Bishop** Equation 8.72), and then marginalising out all the unwanted RVs.
- Make sure you understand that a “message” in BP can be implemented as a factor and can be calculated using factor operations. It is also very useful to attach some intuitive meaning to messages: A message sent from one part of the graph can be understood as what that part of the graph “knows” about the RV of the message. That is why the marginal posterior belief over a certain RV is calculated as the product of all the incoming messages to the variable node for that RV – the marginal belief is the combination of what all the different parts of the graph “knows” about that RV.
- Numerical issues: With unnormalised potentials (as we typically use in MNs) we run a risk of exhausting our numerical range with either under- or overflows. We basically have two avenues to cure this
 - After calculating a message, we simply normalise it (see discussion in **Barber** at the end of Section 5.1.2). This is the approach we will take in this module.
 - We can define and send log messages (see discussion in **Barber** on p. 90 in Section 5.1.4). This formulation has some advantages, but require us to redefine messages. Take note of this possibility, but we will not use log messages in this module.

- For some type of problems, it might be necessary to calculate the partition function or the likelihood of the given evidence (see **Barber** p. 89 in Section 5.1.4). For example, if we want to choose between two different models, the likelihood of the data (i.e., the given evidence) might help us to make this decision. Take note of the idea, but we will not use it in this module (we anyway lose the ability to calculate the partition function or the likelihood of evidence if we normalise messages – see the previous point).
- Make sure that you understand the problem with loops (**Barber** Section 5.1.5). We will look at two approaches to address this problem next week.
- The **PGM Tutorial** discusses the choice of variable elimination ordering in Section 4.1.4. Note that this discussion applies to general graphs; however, we restrict ourselves to models that can be represented as factor trees for now. Take note that choosing an elimination ordering can be a difficult in general, but otherwise you can ignore this section. It should be quite obvious how to choose a good elimination ordering for factor trees.
- Section 4.1.5 in the **PGM Tutorial** is an advanced topic – you can ignore it. Although we will use the idea discussed in this section later (specifically to swap out the marginalisation operation for the maximisation operation to form the max-product algorithm in Week 6), we will not formally describe it or study *commutative semirings*.